

Bioinformatics for engineers

Introduction to Bioinformatics

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FIRST!!!!!!



CONTENTS

1. What is bioinformatics?

2. Remember rememeber the central dogma



3. A brief history of the computers



4. Maggie and her atlas



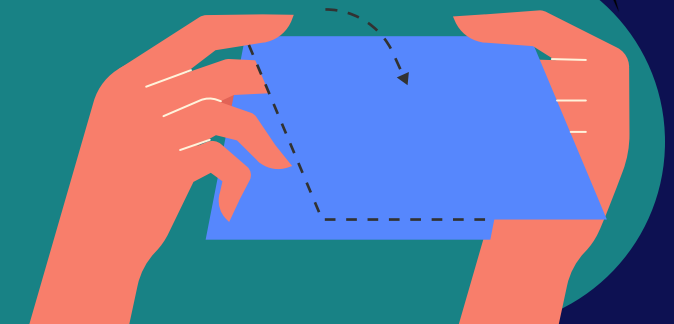
5. The human genome

6. Cheapest is better: next-gen sequencing



7. uses

8. The IAs turn



WHAT IS BIOINFORMATICS?

Bioinformatics or computational biology?



Bioinformatics

Multidisciplinary field that uses computer science tools to store and process biological data.



C. biology

Design and develop algorithms and tools for biological data.



National Institutes of Health
Turning Discovery Into Health

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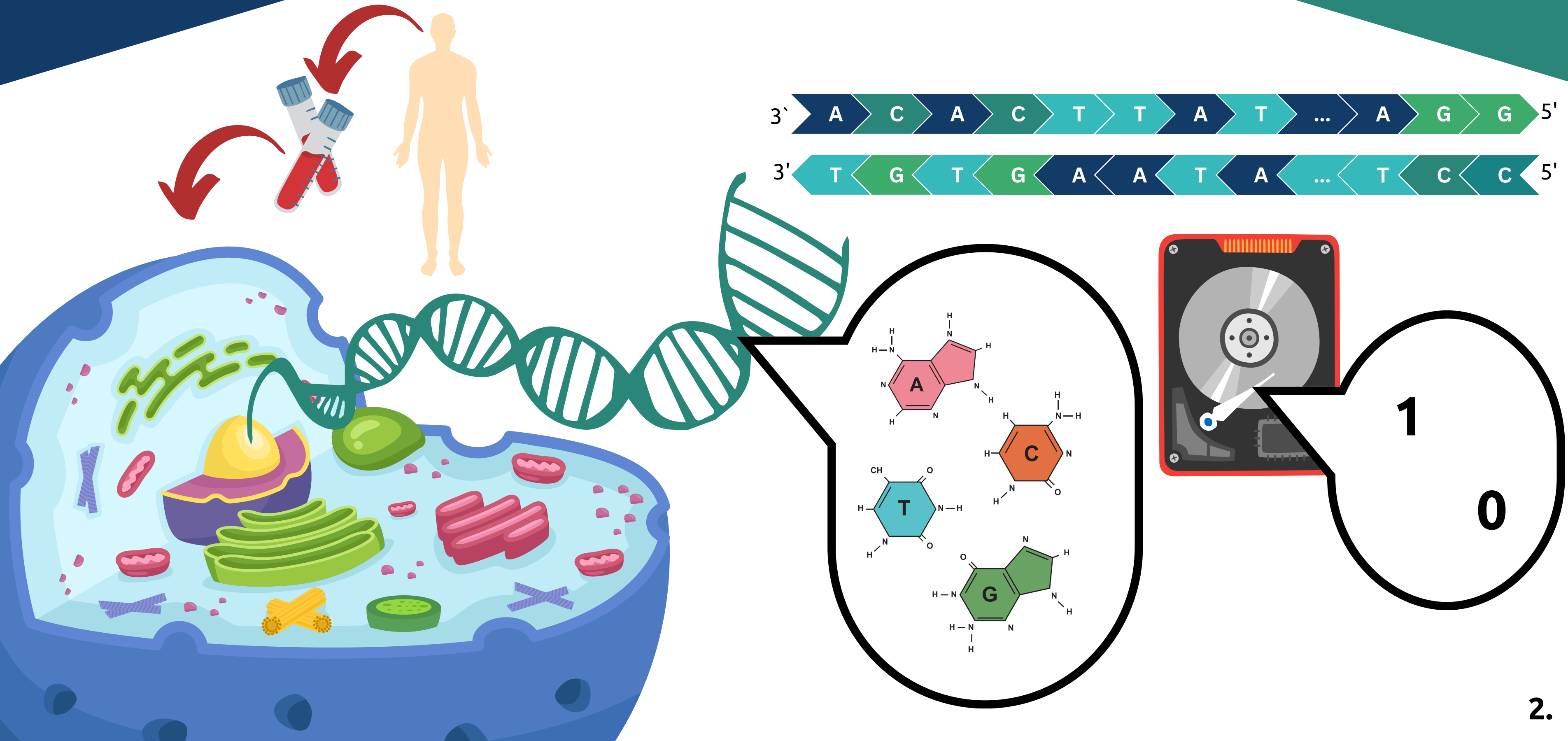


National Institutes of Health
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THE CENTRAL DOGMA

THE GENOME

A live database



3'

5'

Watson and crick AND ROSALIND FRANKLIN!!!

5'

3'

760

NATURE

October 24, 1953 VOL. 172

The X-ray diffraction photograph of crystalline deoxyribonucleic acid (sodium salt) is shown in Fig. 1. In view of its very complex biological functions (for example, the genetical transforming¹ of bacteria) it is remarkable that its crystallinity is so perfect, for such crystallinity indicates a large element of molecular simplicity. We have found no difference in the diffraction patterns of crystalline deoxyribonucleic acid obtained from calf thymus, mouse sarcoma, human white blood cells, *E. coli*, pneumococcus and *Paracentrotus* sperm, although the ratio of bases in the deoxyribonucleic acids varies considerably with species. These facts, combined with the irregularity of sequence of nitrogen bases^{2,3} and the hydrogen bonding between bases^{4,5}, produce a contradiction which is only resolved by a simplifying concept such as the specific pairing of bases suggested by Watson and Crick⁶.

A qualitative view of the X-ray photograph shows several main features which point clearly to the type of structure involved, but the data may be interpreted with more certainty when the reflexions are indexed and intensities measured and plotted in reciprocal space. By using a high-resolution pinhole camera, about a hundred and twenty separate reflexions were resolved in the fibre photograph. Intensities were measured using a microdensitometer. The values are preliminary.

The unit cell of calf thymus deoxyribonucleic acid has been given⁷ as face-centred monoclinic:

$a = 22.0 \text{ \AA}$, $b = 39.8 \text{ \AA}$, $c = 28.1 \text{ \AA}$, $\beta = 96\frac{1}{2}^\circ$.

We find the values for mouse sarcoma deoxyribonucleic acid to be:

$a = 22.2 \text{ \AA}$, $b = 40.4 \text{ \AA}$, $c = 28.1 \text{ \AA}$, $\beta = 97.1^\circ$.

Deduction of the Main Features of the Structure

Fig. 2 shows a two-dimensional view of the intensities of reflexions plotted in reciprocal space. When reflexions overlap, the intensity is divided by the number of reciprocal lattice points possibly involved.

(a) *The structure is helical.* The pattern corresponds in striking fashion with the Fourier transform of a system of helices. The large empty trapezoidal space on and near the meridian strongly suggests a helix but could, however, also be given by alternative structures in which rods or sheets are inclined to the axis. These pseudo-helical structures can be distinguished, however, by the fact that they do not give several zeros of intensity on the layer lines, and the form function is not circularly symmetrical about the fibre axis. If paracrystalline deoxyribonucleic acid only were available, it would be impossible to establish the circular symmetry of the form function,

and hence three-dimensional data obtained from study of the crystalline acid are necessary if the helical nature of the structure is to be proved. This is discussed later.

(b) *A major part of the helix has one sharply defined diameter.* The intensity distribution on the second layer line and on the outer parts of the equator alternates markedly between strong and weak and corresponds respectively with the functions J_0^2 and J_2^2 (squares of second- and zero-order Bessel functions). This diffraction would be produced by a helix of sharply defined diameter of about 18 Å. The intensity on the 4th, 5th, 6th, 7th, 9th and 11th layer lines corresponds roughly to the main maxima of the functions J_2^2 , J_4^2 , J_6^2 , J_8^2 , J_{10}^2 and J_{12}^2 (Fig. 2), again for the same diameter of 18 Å.

(c) *There are two coaxial 18 Å. diameter helices spaced half a pitch-length apart.* The 1st and 3rd layer lines are weaker than the second, and the intensities correspond roughly with J_1^2 and J_3^2 (Fig. 2) respectively, for a helix of diameter approximately 10 Å. The strong diffraction from 18 Å. diameter helices observed on the 2nd layer line, but absent from 1st and 3rd, indicates that the structure contains two 18 Å. diameter coaxial helices spaced apart along the fibre axis by about half the pitch-length of 28 Å. (the layer line spacing) (see also ref. 8). The inner regions of the equator correspond to a

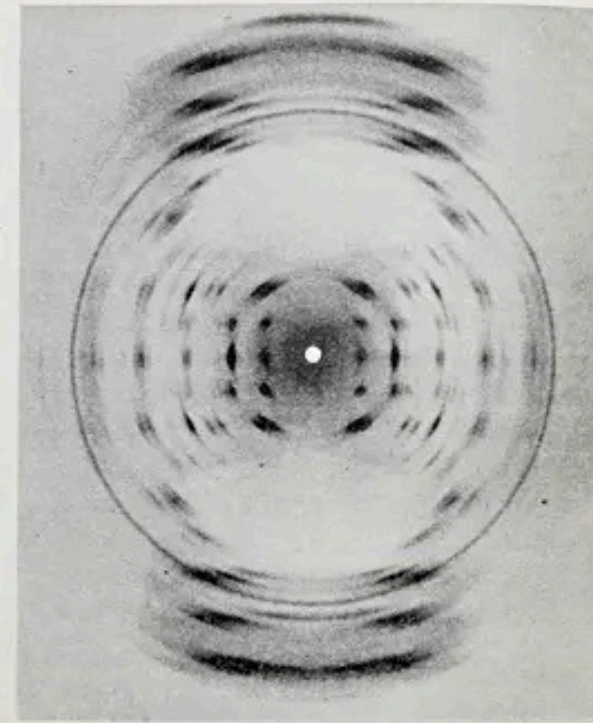


Fig. 1. X-ray diffraction photograph of a bundle of four fibres of crystalline mouse sarcoma deoxyribonucleic acid (sodium salt) at approximately 75 per cent relative humidity. The ring is due to impurity. (Deoxyribonucleic acid supplied by R. Barclay and L. D. Hamilton)

No. 4382

October 24, 1953

NATURE

761

smaller average diameter than 18 Å., and if the 10 Å. helix extends over a range of diameter from about 9 to 14 Å. it will contribute mainly to the inner regions of the equator and will not affect much the outer region, which corresponds to J_0^2 of 18 Å. The equatorial intensity distribution will then be accounted for by the addition of the amplitudes diffracted by the 18 and 10 Å. helices (Fig. 2). The 18 Å. diameter helices do not contribute J_0^2 to the 4th layer line; hence the thickness of the 18 Å. helix in the axial direction must be at least 3.5 Å.

(d) *Spaced centrally between the two 18 Å. diameter helices is one helix of mean diameter about 10 Å.* We expect that the 10 Å. helix may contribute to the intensity of the 2nd layer line. If this intensity followed accurately a J_2^2 function, the heights of the maxima first and second closest to the centre of the layer line would be in the ratio 1:0.4. In fact, these heights are about equal. If a small amplitude of J_0 , corresponding to 10 Å. diameter, is added out of phase to the 18 Å. J_0 , such an effect would be produced (Fig. 2). Hence we expect the 10 Å. diameter helix to be spaced along the helix axis halfway between the two 18 Å. diameter helices, the whole system being coaxial. This 10 Å. diameter helix is apparently that part of the structure which Franklin and Gosling have remarked⁹ is not repeated 14 Å. apart along the fibre axis. Indications of this single helix can, in fact, be seen on their two-dimensional Patterson diagram.

(e) *There are eleven nucleotides per turn of one helix.* This is deduced from the fact that there is a system of Bessel function maxima converging on the centre of the 11th layer line (see also ref. 8).

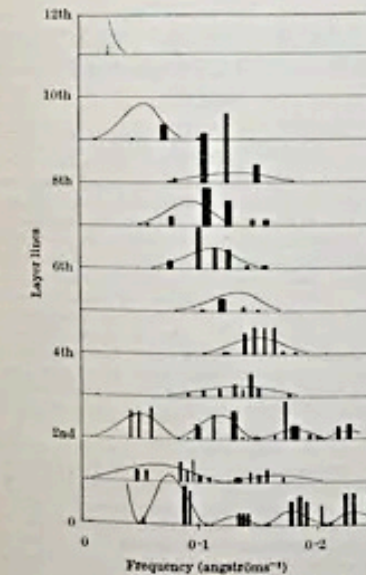


Fig. 2. Plot in reciprocal space of X-ray reflexions from crystalline deoxyribonucleic acid; fibre axis vertical. The height of each black rectangle is proportional to the observed form factor of the structure, and the area to the total intensity produced by overlapping reflexions. The continuous curve corresponds to the roughly calculated form factor for the helical structure suggested in the text.

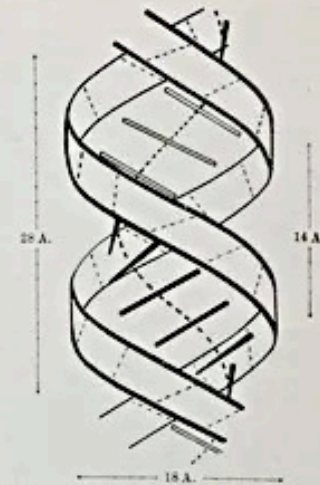


Fig. 3. The type of structure of the helical unit deduced from X-ray data of crystalline deoxyribonucleic acid. One helix of diameter approximately 10 Å. is formed by a system of eleven inclined rods per turn of the helix. This helix is surrounded by two 18 Å. diameter helices, each of which is also formed of eleven units per turn.

(f) *The nucleotide shape resembles that of a rod inclined to the helix axis.* If the helical system is divided into nucleotides by planes at right angles to the helix axis, the intensities of the 10th, 9th and 8th layer lines would resemble that of the zero, 1st, 2nd and 3rd layer lines. In fact, this effect is observed in so far as the 18 Å. diameter helix does contribute to the 9th layer line (though weakly) and is partly absent from the 10th. But it is anomalous that the 10 Å. diameter helix does not contribute to the 10th layer line and the 18 Å. helix does contribute, as J_1^2 , to the 7th layer line. These effects would be explained if the nucleotides formed roughly a series of rods inclined at an angle of about 65° to the fibre axis in the opposite sense to the inclination of the helix. The X-rays are then reflected off these rods in a direction about 25° to the fibre axis. The 10 Å. diameter helix, being in effect divided by inclined planes, diffracts very little energy along the meridian, and the 10th layer line is accordingly absent.

The structure of the helical system therefore begins to become clear and its probable form is shown diagrammatically in Fig. 3.

Circular Symmetry of the Form Function and the Elimination of Pseudo-Helical Structures

The fact that the form function, for example, on the 2nd layer line, alternates fairly smoothly between strong and very weak, indicates that it has circular symmetry about the fibre axis. Hence all pseudo-helical structures are eliminated. However, as the lattice is asymmetrical, a symmetrical form function is quite likely to produce general asymmetry of the intensity distribution. This effect, combined with some ambiguity of indexing, gave rise to the suggestion⁹ that there might be a very definite asymmetry in the form function and therefore in the structure itself. If this were the case, the structure could not be helical unless the helix were considerably distorted. Probably the greatest asymmetry of reflected energy

3'

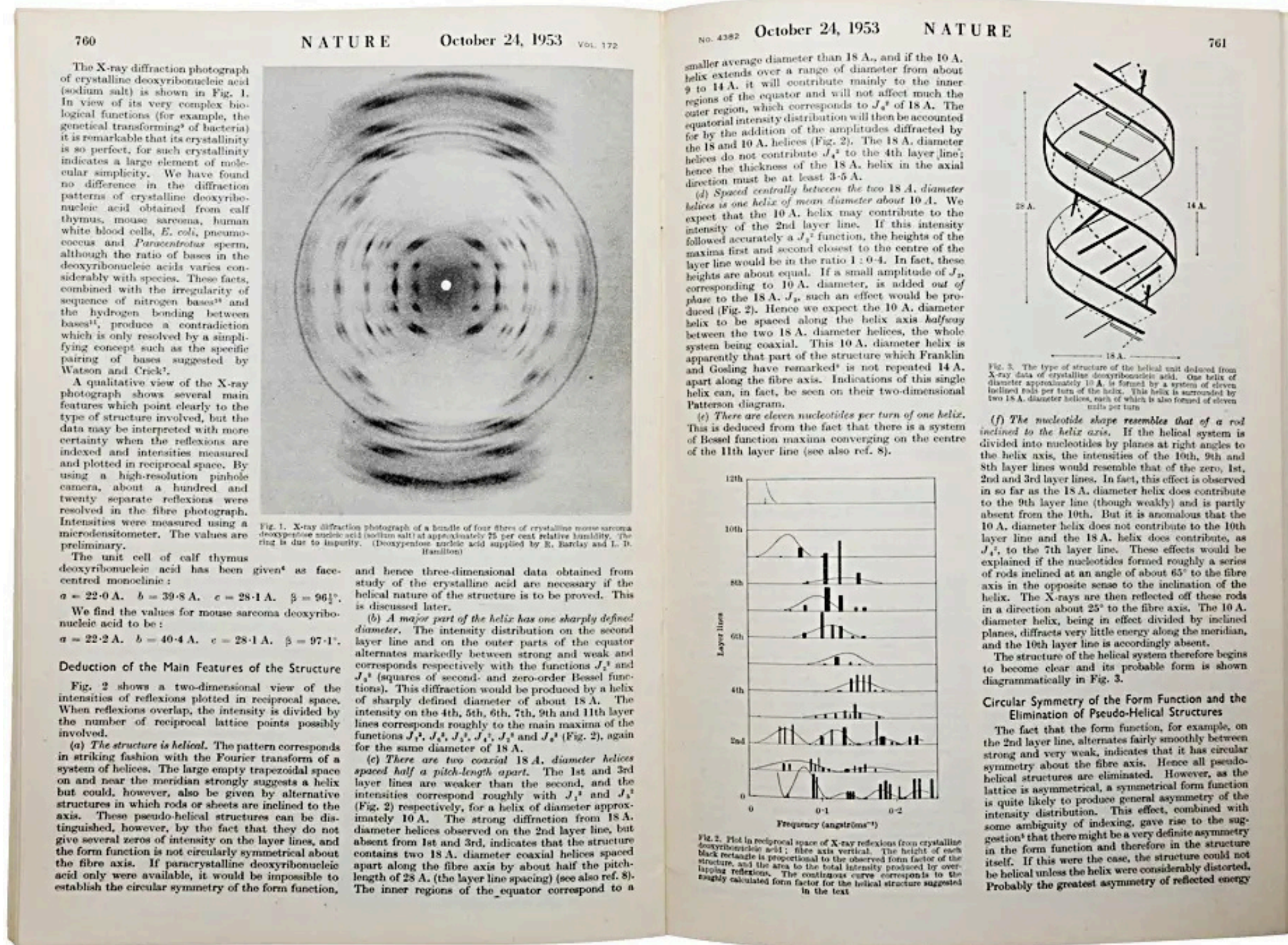
5'

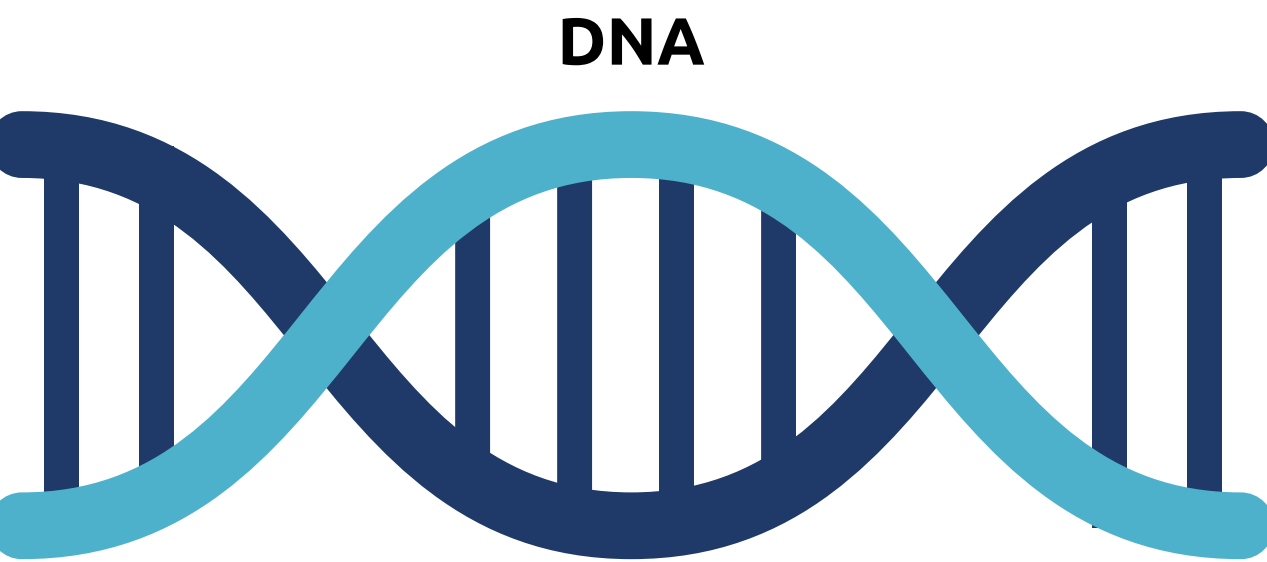
Watson and crick AND ROSALIND FRANKLIN!!!

the meaning of life
is 5' to 3'.

5'

3'





DNA

CENTRAL DOGMA

a biological decompression algorithm

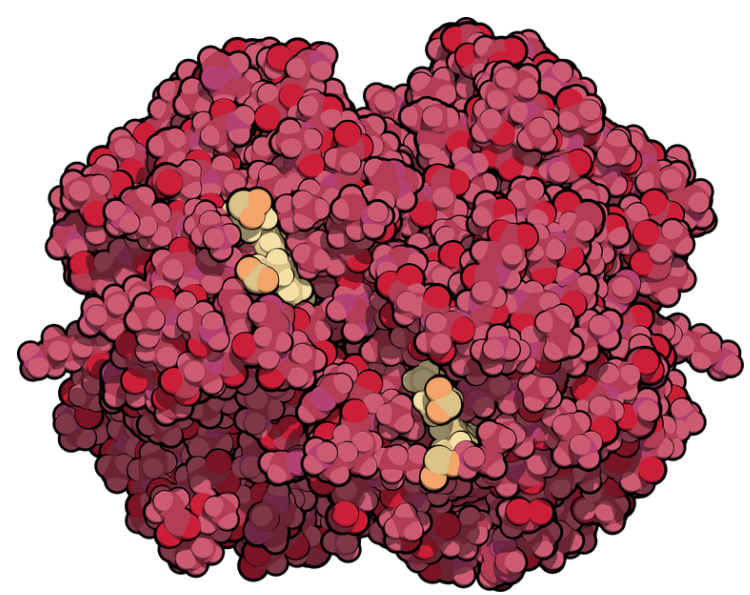
Gene

3' A C A C T T A T ... A G G 5'

3' T G T G A A

RNA_m

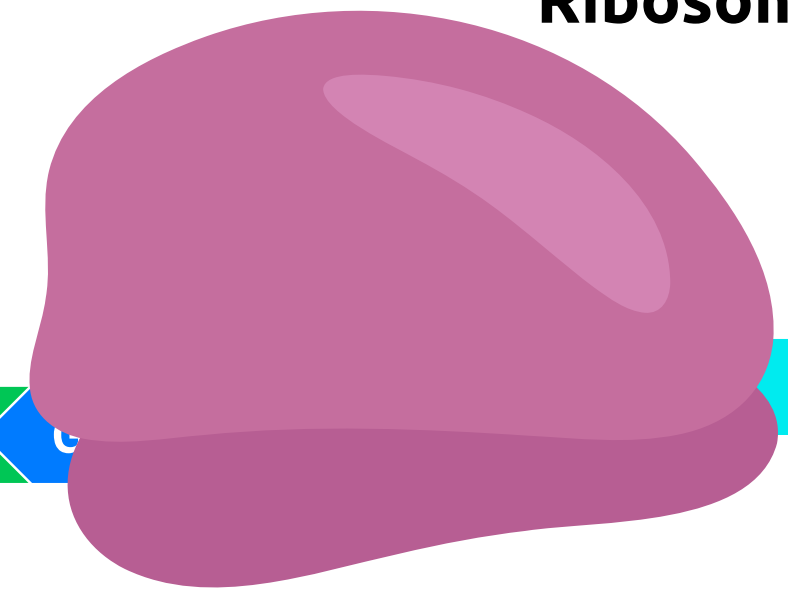
U A .. U C C



Protein



Ribosome

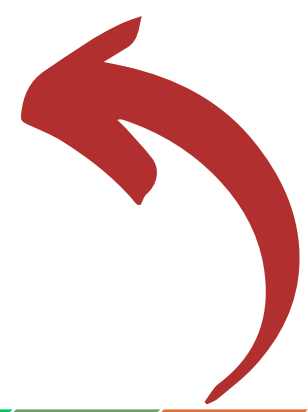


RNA_m

U U C C

F L M T A L C

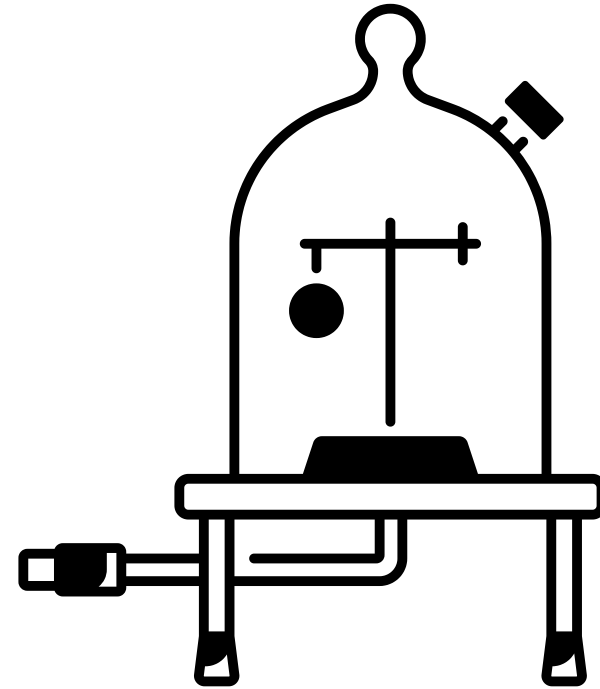
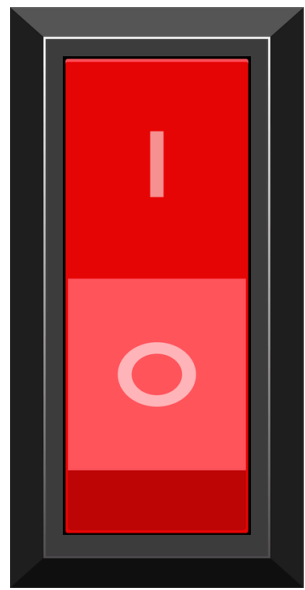
AA Seq



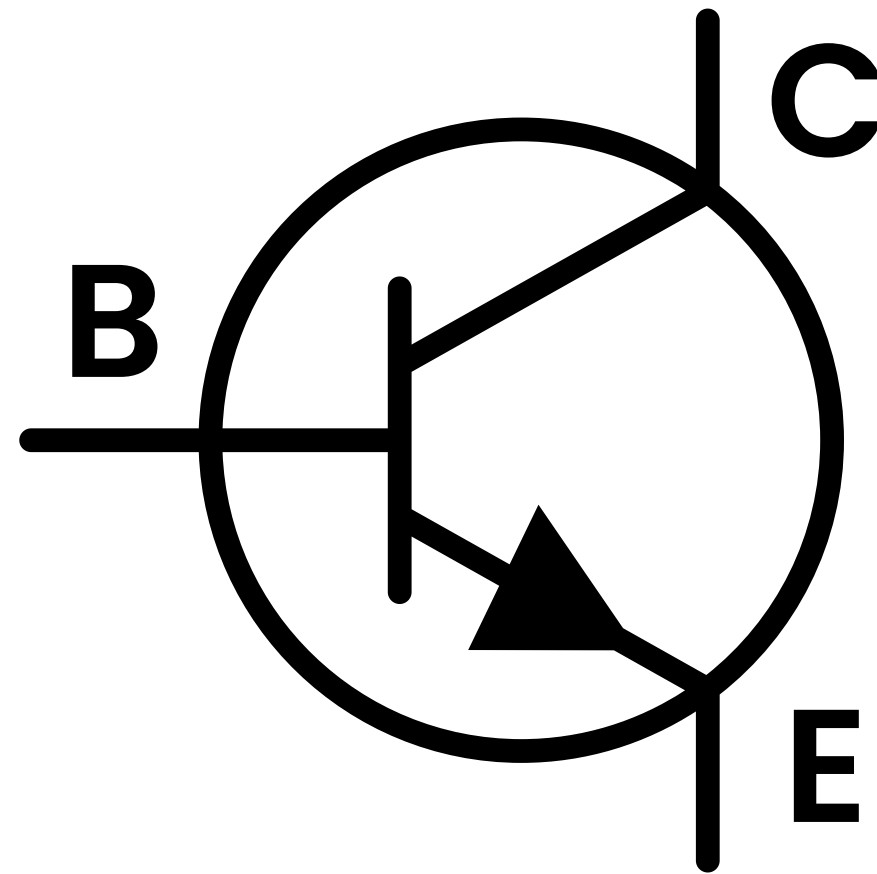
THE COMPUTER SCIENCE HISTORY

Transistors

We need to represent a binary system

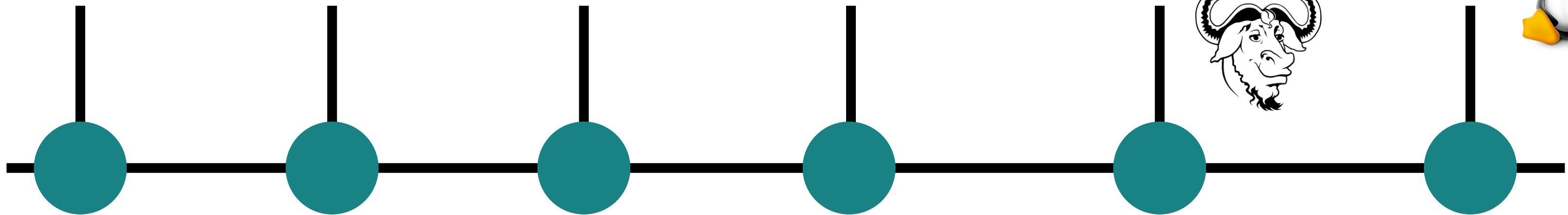
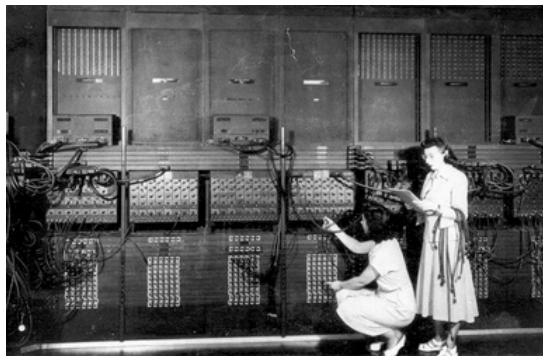


Thermoionization controlled by induced current.



An electronic bridge!!!!!!!

Computer timeline



1945

ENIAC, the first electronic computer used to calculate artillery tables

1951

UNIVAC, first-generation computer based on vacuum tubes

1959

Second-generation computer (IBM) based on transistors

1969

UNIX borns on Dennis Ritchie's hands

1983

GNU project

1991

linux kernel

THE FIRST PATTERNS IN A SEQUENCE

THE PROTEIN ATLAS



Substitution table based ON 65 KNOWN PROTEINS
(1965)

Conserved domains may have
the same functions !!!!!



Margaret O. Dayhoff

	A	R	N	D	C	Q	E	G	H	I	L	K	M	F	P	S	T	W	Y	V
A	13	6	9	9	5	8	9	12	6	8	6	7	7	4	11	11	11	2	4	9
R	3	17	4	3	2	5	3	2	6	3	2	9	4	1	4	4	3	7	2	2
N	4	4	6	7	2	5	6	4	6	3	2	5	3	2	4	5	4	2	3	3
D	5	4	8	11	1	7	10	5	6	3	2	5	3	1	4	5	5	1	2	3
C	2	1	1	1	52	1	1	2	2	2	1	1	1	1	2	3	2	1	4	2
Q	3	5	5	6	1	10	7	3	7	2	3	5	3	1	4	3	3	1	2	3
E	5	4	7	11	1	9	12	5	6	3	2	5	3	1	4	5	5	1	2	3
G	12	5	10	10	4	7	9	27	5	5	4	6	5	3	8	11	9	2	3	7
H	2	5	5	4	2	7	4	2	15	2	2	3	2	2	3	3	2	2	3	2
I	3	2	2	2	2	2	2	2	2	10	6	2	6	5	2	3	4	1	3	9
L	6	4	4	3	2	6	4	3	5	15	34	4	20	13	5	4	6	6	7	13
K	6	18	10	8	2	10	8	5	8	5	4	24	9	2	6	8	8	4	3	5
M	1	1	1	1	0	1	1	1	1	2	3	2	6	2	1	1	1	1	1	2
F	2	1	2	1	1	1	1	1	3	5	6	1	4	32	1	2	2	4	20	3
P	7	5	5	4	3	5	4	5	5	3	3	4	3	2	20	6	5	1	2	4
S	9	6	8	7	7	6	7	9	6	5	4	7	5	3	9	10	9	4	4	6
T	8	5	6	6	4	5	5	6	4	6	4	6	5	3	6	8	11	2	3	6
W	0	2	0	0	0	0	0	0	1	0	1	0	0	1	0	1	0	55	1	0
Y	1	1	2	1	3	1	1	1	3	2	2	1	2	15	1	2	2	3	31	2
V	7	4	4	4	4	4	4	5	4	15	10	4	10	5	5	5	7	2	4	17

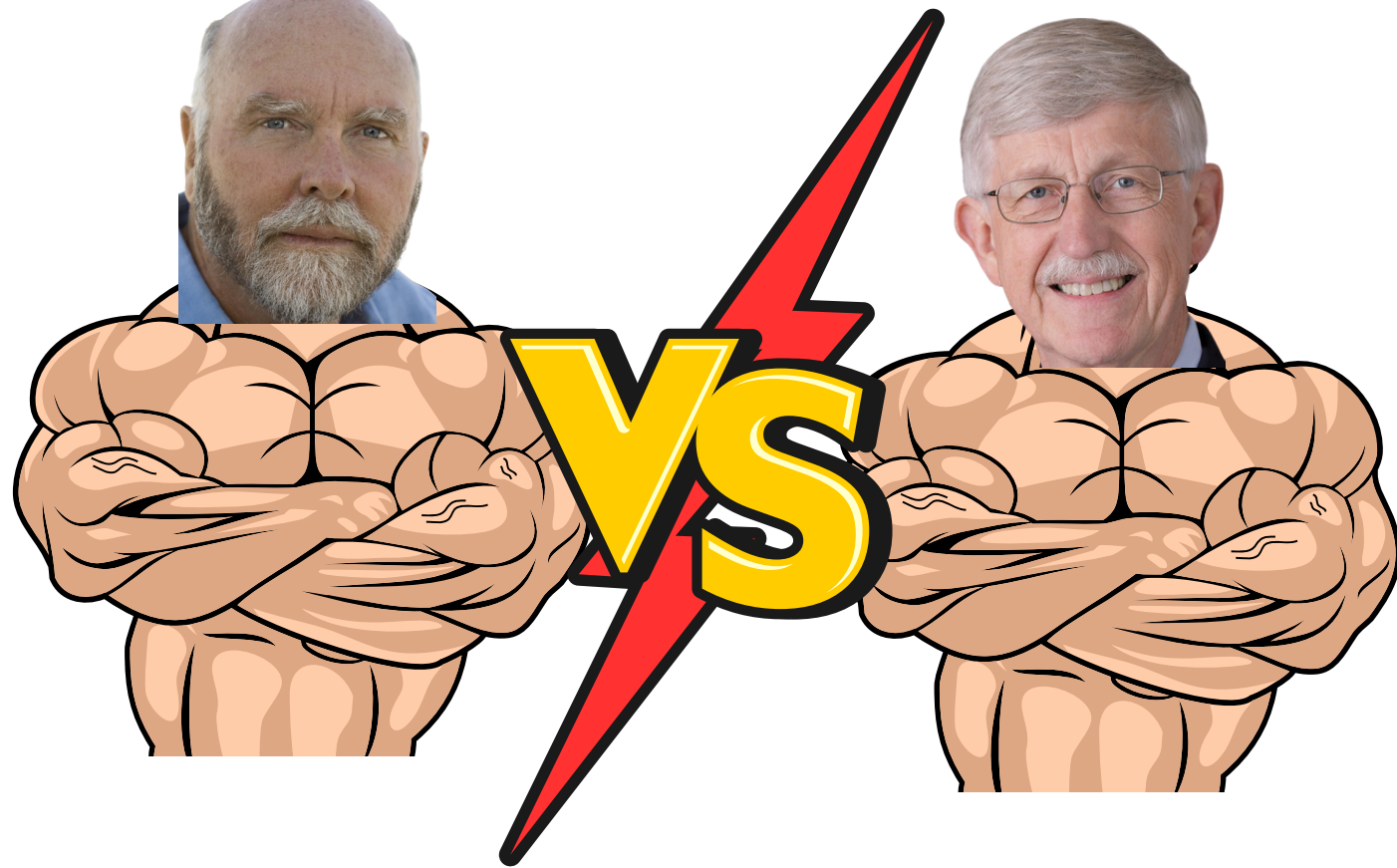
THE HUMAN GENOME PROJECT

Collins vs Venter



Craig venter

Francis Collins



born in 1946, founder of celera genomic, where he tried to obtain the human genome.

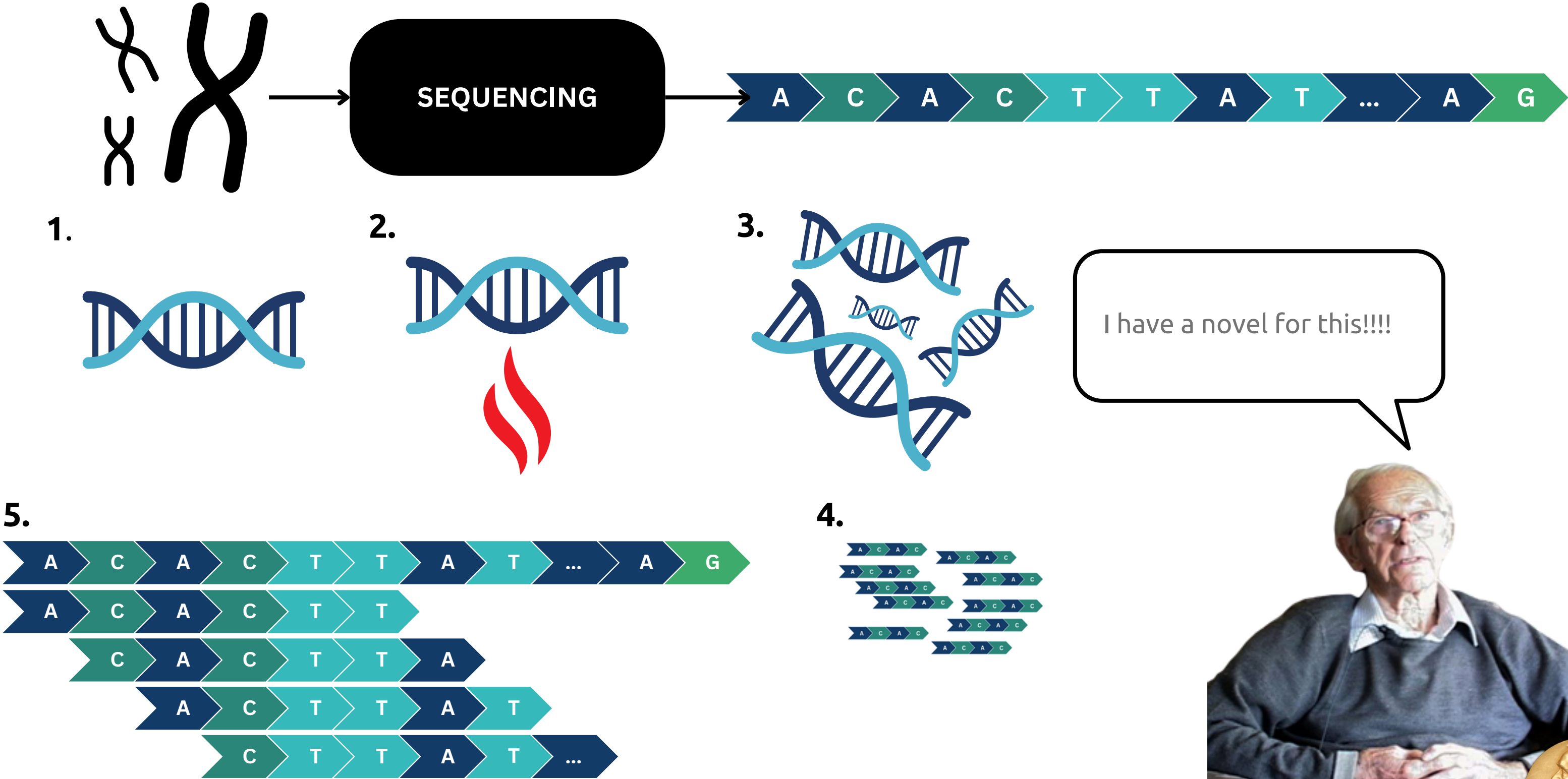
Born in 1950, Collins was a geneticist that led the human genome project.

- Venter intended to patent the genome or at least parts of it, while collins relied on open cycling.
- Collins used regions mapped during sequencing, Venter intended to use the shotgun method.
- The battle accelerated the scientific process given the competition.
- collins published in nature and venter in science, within days of each other.

**HOW TO OBTAIN THE
SEQUENCE?**

GENOME SECUENCING

how to read thiS biological book?



I have a novel for this!!!!



Frederick Sanger

SOME CASES OF USE

The map of use

Bioinformatics

Sequence reconstruction

reconstructing long
sequences from partial
information sections

Annotation

Understand the structure
of the sequences and
their subsections.

Functional analysis

Determine the
relationship of genetic
information to
phenotype

Comprative studies

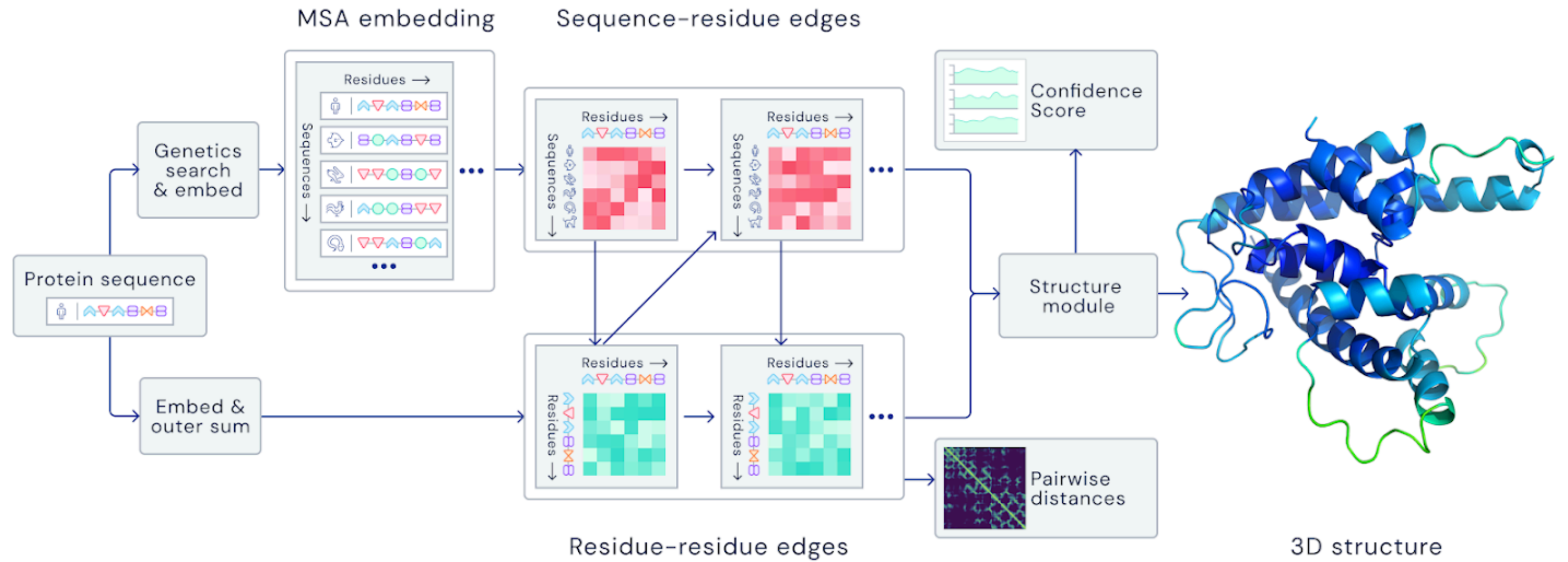
To understand the
relationship between
different organisms.

Lab optimization

Construction of tools to
facilitate the laboratory
processes.

**MACHINES ARE TAKING
OUR JOBS!!!!**

Protein folding



Demis Hassabis

Thanks !!!!!